Git is a distributed version control system widely used for source code management. It has numerous parameters and options that can be configured to manage repositories effectively. Below is a comprehensive list of common Git parameters along with their definitions and descriptions.

General Git Parameters

Configuration Parameters (`git config`)

- **`user.name`**
  - **Definition**: The name used in commits.
  - **Description**: Sets the user name for commits.

- **`user.email`**
  - **Definition**: The email address used in commits.
  - **Description**: Sets the email address for commits.

- **`core.editor`**
  - **Definition**: The text editor used by Git.
  - **Description**: Specifies the default editor for commit messages and other text files.

- **`core.autocrlf`**
  - **Definition**: Auto-converts line endings.
  - **Description**: Ensures consistent line endings across different operating systems.

- **`remote.origin.url`**
  - **Definition**: URL of the remote repository.
  - **Description**: Sets the URL for the `origin` remote.

- **`branch.<name>.merge`**
  - **Definition**: The remote branch to track.

- **Description**: Specifies the branch to merge from a remote.

- **`alias.<name>`**
  - **Definition**: Defines a Git command alias.
  - **Description**: Creates shortcuts for Git commands.

Initialization Parameters (`git init`)

- **`--bare`**
  - **Definition**: Initializes a bare repository.
  - **Description**: Creates a repository without a working directory.

- **`--template`**
  - **Definition**: Specifies a directory from which templates will be copied.
  - **Description**: Uses templates from the given directory.

- **`--separate-git-dir`**
  - **Definition**: Specifies a separate directory for the `.git` folder.
  - **Description**: Places the `.git` directory in a specified location.

Common Git Commands and Their Parameters

`git clone`

- **`--branch`**
  - **Definition**: Specifies a branch to clone.
  - **Description**: Clones a specific branch instead of the default branch.

- **`--depth`**
  - **Definition**: Creates a shallow clone with a history truncated to the specified number of commits.

  - **Description**: Reduces the amount of data cloned.

- **`--single-branch`**
  - **Definition**: Clones only one branch.
  - **Description**: Does not fetch data for other branches.

- **`--recurse-submodules`**
  - **Definition**: Clones submodules within the repository.
  - **Description**: Fetches and checks out submodules.

`git add`

- **`--all` or `-A`**
  - **Definition**: Adds all changes.
  - **Description**: Stages all changes (new, modified, and deleted files).

- **`--patch` or `-p`**
  - **Definition**: Interactively adds changes.
  - **Description**: Allows you to choose hunks of changes to add.

- **`--force` or `-f`**
  - **Definition**: Forces the addition of files.
  - **Description**: Adds files even if they are ignored by `.gitignore`.

`git commit`

- **`--message` or `-m`**
  - **Definition**: Specifies the commit message.
  - **Description**: Sets the commit message directly from the command line.

- **`--all` or `-a`**
  - **Definition**: Automatically stages files that have been modified and deleted.
  - **Description**: Skips `git add` for modified and deleted files.

- **`--amend`**
  - **Definition**: Modifies the most recent commit.
  - **Description**: Allows you to edit the last commit.

- **`--signoff` or `-s`**
  - **Definition**: Adds a `Signed-off-by` line at the end of the commit message.
  - **Description**: Indicates who is responsible for the commit.

`git pull`

- **`--rebase`**
  - **Definition**: Rebase the current branch on top of the upstream branch after fetching.
  - **Description**: Avoids unnecessary merge commits.

- **`--no-commit`**
  - **Definition**: Does not commit changes after merging.
  - **Description**: Leaves changes uncommitted.

- **`--ff-only`**
  - **Definition**: Allows only fast-forward merges.
  - **Description**: Fails if fast-forward is not possible.

`git push`

- **`--force` or `-f`**
  - **Definition**: Forces the push.

- **Description**: Overwrites the remote branch with the local branch.

- **`--set-upstream` or `-u`**
  - **Definition**: Sets the upstream branch.
  - **Description**: Sets the remote branch to track.

- **`--tags`**
  - **Definition**: Pushes tags to the remote repository.
  - **Description**: Includes all tags in the push.

`git branch`

- **`--all` or `-a`**
  - **Definition**: Lists all branches.
  - **Description**: Shows both local and remote branches.

- **`--delete` or `-d`**
  - **Definition**: Deletes a branch.
  - **Description**: Removes the specified branch.

- **`--move` or `-m`**
  - **Definition**: Renames a branch.
  - **Description**: Changes the branch name.

`git merge`

- **`--no-ff`**
  - **Definition**: Creates a merge commit even if the merge could be resolved with a fast-forward.
  - **Description**: Ensures that a merge commit is created.

- **`--squash`**
  - **Definition**: Squashes commits into a single commit.
  - **Description**: Combines changes from the feature branch without committing.

- **`--abort`**
  - **Definition**: Aborts the merge process.
  - **Description**: Resets the state to before the merge began.

`git rebase`

- **`--interactive` or `-i`**
  - **Definition**: Rebase interactively.
  - **Description**: Allows editing commits, squashing, and reordering.

- **`--onto`**
  - **Definition**: Specifies the new base for the rebased branch.
  - **Description**: Rebases the current branch onto the specified commit.

- **`--continue`**
  - **Definition**: Continues the rebase process after resolving conflicts.
  - **Description**: Proceeds with the rebase.

- **`--abort`**
  - **Definition**: Aborts the rebase process.
  - **Description**: Resets the state to before the rebase began.

`git log`

- **`--oneline`**
  - **Definition**: Displays each commit on a single line.

  - **Description**: Provides a brief summary of the commit history.

- **`--graph`**
  - **Definition**: Displays an ASCII graph of the branch and merge history.
  - **Description**: Visualizes the commit history.

- **`--decorate`**
  - **Definition**: Adds branch and tag names to the commit logs.
  - **Description**: Shows references alongside commit messages.

- **`--author`**
  - **Definition**: Shows commits by a specific author.
  - **Description**: Filters commits by the author's name.

- **`--since`**
  - **Definition**: Shows commits more recent than a specific date.
  - **Description**: Filters commits by date.

- **`--until`**
  - **Definition**: Shows commits older than a specific date.
  - **Description**: Filters commits by date.

`git diff`

- **`--staged` or `--cached`**
  - **Definition**: Shows changes between the index and the last commit.
  - **Description**: Compares staged changes with the latest commit.

- **`--name-only`**
  - **Definition**: Shows only names of changed files.

- **Description**: Lists files that have changed.

- **`--name-status`**
  - **Definition**: Shows names and status of changed files.
  - **Description**: Provides a summary of changes (e.g., added, modified, deleted).

- **`--stat`**
  - **Definition**: Displays a summary of changes.
  - **Description**: Shows file names, the number of insertions and deletions, and a summary line.

`git status`

- **`--short` or `-s`**
  - **Definition**: Displays a compact summary of the status.
  - **Description**: Provides a brief overview of changes.

- **`--branch` or `-b`**
  - **Definition**: Shows the branch and tracking information.
  - **Description**: Includes the branch name and upstream tracking branch.

- **`--ignored`**
  - **Definition**: Shows ignored files.
  - **Description**: Lists files that are ignored by `.gitignore`.

This list covers many of the essential Git parameters. Specific Git commands and configurations may introduce additional parameters and options. Always refer to the [Git documentation](https://git-scm.com/doc) for the most

comprehensive and up-to-date information.